INCH-POUND
MIL-DTL-0048683B (AR)
14 April 2004
USED IN LIEU OF
MIL-C-48683A (AR)
26 October 1987

DETAIL SPECIFICATION

CARTRIDGE 40MM PRACTICE - TP M918- PROJECTILE ASSEMBLY FOR METAL PARTS AND LOADING, ASSEMBLING AND PACKAGING

This specification is approved for interim use by the U.S. Army Armament Research, Development and Engineering Center (ARDEC). Other activities in the Department of Defense may use this interim revision or may continue using MIL-C-48683A (AR).

1. SCOPE

1.1 Scope. This specification covers the metal parts and loading, assembling and packaging for one type of projectile assembly for use in the Cartridge, 40MM: TP, M918.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless specified, the issues of these documents are those cited in the solicitation .

Comments, suggestions, or questions on this document should be addressed to: Commander, US Army ARDEC, ATTN: AMSRD-AAR-QES-E, Picatinny, New Jersey 07806-5000, or e-mailed to ardec-stdzn@pica.army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil.

AMSC N/A FSC 1310
DISRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:36Z
Check the source to verify that this is the current version before use.

FEDERAL SPECIFICATIONS

TT-C-490	Chemical Conversion Coatings and Pre-Treatments for Ferrous Surfaces (Base for Organic Coatings)

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-A-8625	Anodic Coating, for Aluminum and Aluminum Alloys
MIL-A-70625	Automated Acceptance Inspection Equipment Design, Testing and Approval of

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-331	Fuze and Fuze Components Environmental and Performance Tests For
MIL-STD-1168	Ammunition Lot Numbering and Ammunition Data Card
MIL-STD-1916	DOD Preferred Method for Acceptance of Product
MIL-STD-2073-1	DOD Standard Practice for Military Packaging

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation (see 6.2).

US ARMY ARDEC DRAWINGS (see 6.7)

8886327 - M169, 40MM Cartridge Case Assembly
8886357 - Escapement Assembly
9399372 - Cartridge, 40MM: M918 TP
9399374 - Projectile Assembly
9399382 - Body Insert
9399389 - Flash Charge Composition, bulk, prior to loading into capsule
9399390 - Ogive Assembly, 40MM: M918
9399391 - Firing Pin Assembly
9399392 - Firing Pin
9399393 - Firing Pin Plate
9399394 - Ogive M918TP
9399396 - Body
9399402 - Body Assembly
9399405 - Cap, Inner
9399406 - Cap, Outer
9399407 - Capsule, Assembly

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:36Z
Check the source to verify that this is the current version before use.

9399408 - Spring, Anti-creep
12624588 - Packing and Marking for Box, Fiberboard, Outer, for projectile, 40MM: M918 TP

(Copies of these drawings may be requested online at drawing-request@pica.army.mil, or from U.S. Army ARDEC, ATTN: AMSRD-AAR-AIS-TD, Picatinny NJ 07806-5000.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR TESTING AND MATERIALS INTERNATIONAL (ASTM)

ASTM B117 - Standard Method of Salt Spray (fog) Testing
ASTM C208 - Standard Specification For Cellulose Fiber Insulating Board

(Copies of these documents are available from www.astm.org or ASTM International, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the solicitation or contract order (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.3.

3.2 Materials. Materials, parts, and assemblies shall comply with requirements specified on the applicable drawings and referenced specifications.

3.3 Non-Functioning. The projectile shall not lodge in bore of the weapon and there shall be no evidence of a premature burst (see 6.6.2).

3.4 Functioning. The projectile assembly, on impact, shall produce sound, smoke and flash under standard atmospheric conditions. There shall be no evidence of an early burst (see 6.6.1) or premature burst (6.6.2). For 1,000 meter range, sound and smoke or sound and flash are also acceptable.

3.4.1 Safety. The projectile shall not lodge in the bore of the weapon and there shall be no evidence of a premature burst (see 6.6.2). There shall be no evidence missing petals or blown base.

3.5 Ammunition lot numbering. Ammunition lot numbers shall be assigned in accordance with MIL-STD-1168.

3.6 Jolt. The escapement assembly, assembled to a projectile, shall be x-rayed

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:36Z
Check the source to verify that this is the current version before use.

and shall not be armed or partially armed (as evidenced by the detonator moving any degree from safe position) or have any safety feature missing or not in its proper position (i.e. detent, detent spring, setback pin, setback spring), damaged parts or foreign material. After being subjected to the jolt test, the projectile assembly shall be x-rayed again and shall not be armed or partially armed (as evidenced by the detonator moving any degree from the safe position) or have any safety feature not in its proper position, damaged parts or foreign material.

3.7 Jumble. The escapement assembly, assembled to a projectile shall be x-rayed and shall not be armed or partially armed (as evidenced by the detonator moving any degree from the safe position) or have any safety feature missing or not in its proper position (i.e. detent, detent spring, setback pin, setback spring), damaged parts or foreign material. After being subjected to the jumble test, the projectile assembly shall be x-rayed again and shall not be armed or partially armed (as evidenced by the detonator moving any degrees from the safe position) or have any safety feature no in its proper position, damaged parts or foreign material.

3.8 Transportation vibration. Prior to transportation vibration test, the escapement assembly, assembled to a projectile shall be x-rayed and shall not be armed or partially armed (as evidenced by the detonator moving any degree from the safe position) or have any safety feature missing or not in its proper position (i.e. detent, detent spring, setback pin, setback spring), damaged parts or foreign material. After performance of the transportation vibration test the escapement assembly shall be x-rayed again and shall not be armed or partially armed (as evidenced by the detonator moving any degree from the safe position) or have any safety feature not in its proper position, damaged parts or foreign material and shall meet the requirements of functioning (3.3) and non-functioning (3.4).

3.9 Workmanship. All parts and assemblies shall be fabricated and finished in a thorough and workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, unblended radii, surface defects, dirt, grease, rust, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any parts, nor shall the parts be contaminated by the cleaning agents. All marking and stamping shall be neat and sharply defined.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:36Z
Check the source to verify that this is the current version before use.

4. VERIFICATION

TABLE I. Requirements/Inspection Cross Reference Matrix

METHOD OF INSPECTION
N/A - Not applicable
1 - Analysis
2 - Demonstration
3 - Examination
4 - Test

CLASSES OF INSPECTION
A - First article
B - Conformance

Section 3 Requirement	Inspection Methods					Inspection Class		Section 4 Inspection Method
	N/A	1	2	3	4	A	B	
3.1			X	X	X	X		4.3
3.2			X	X	X	X	X	4.4
3.3				X	X	X	X	4.5.7
3.4				X	X	X	X	4.5.6, 4.5.6.1, 4.5.6.2, 4.5.6.3
3.4.1				X	X	X	X	4.5.6, 4.5.6.1, 4.5.6.2, 4.5.6.3
3.5				X		X	X	4.5.12
3.6				X	X	X	X	4.5.13
3.7				X	X	X	X	4.5.14
3.8				X	X	X	X	4.5.15
3.9				X		X	X	4.5.17

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)
b. Conformance inspection (see 4.4)

4.2 Inspection conditions. Unless otherwise specified, all inspection shall be performed in accordance with the test conditions specified in section 4.5.

4.3 First article verification. When specified in the solicitation or contract, a sample shall be subjected to first article inspection in accordance with Table II.

4.3.1 First article quantity. The contractor shall submit a first article as designated by the Contracting Officer for evaluation in accordance with the provision of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Parts description	Drawings	Quantity
Projectile Assembly	9399374	838
Body Insert	9399382	25
Ogive Assembly, 40MM: M918	9399390	50
Firing Pin Assembly	9399391	50
Firing Pin	9399392	25

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:36Z
Check the source to verify that this is the current version before use.

Firing Pin Plate	9399393	25
Ogive M918TP	9399394	25
Spring, Anti-creep	9399408	50
Body	9399396	25
Body Assembly (prior to painting)	9399402	25
Body Assembly (after to painting)	9399402	25
Cap, Inner	9399405	25
Cap, Outer	9399406	25
Capsule, Assembly	9399407	50

4.3.2 Inspections to be performed. The first article inspection shall be 100% examination for defects of all drawing characteristics (listed and unlisted) including workmanship requirements for the parts and assemblies listed in 4.3.1 and the tests specified in Table II herein.

4.3.3 First article rejection. If any item of the sample fails to comply with the first article requirements, the sample shall be rejected. The Government reserves the right to terminate inspection upon any failure of an assembly, component or test specimen to comply with any of the requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:36Z
Check the source to verify that this is the current version before use.

TABLE II. First Article Inspection.

TITLE: Projectile and Components	SHEET 1 OF 2		DRAWING NUMBER: See Below NEXT HIGHER ASSEMBLY
EXAMINATION OR TEST	N0. OF SAMPLE UNITS	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Firing Pin Assembly (Dwg. 9399391)			
Push Test, Firing Pin Assembly	50	3.2	4.5.1
Ogive, M918 TP (Dwg. 9399394)			
Salt Spray	4	3.2	4.5.2.1
Spring, Anti-Creep (Dwg. 9399408)			
Load Test	50	3.2	4.5.3
Capsule Assembly (Dwg. 9399407)			
Moisture content	50 10/	3.2	4.5.8
Body Assembly (after painting) (Dwg. 9399402)			
Salt Spray	4		4.5.2.2
Push test of rotating band 7/	50	3.2	4.5.10
Projectile Assembly (Dwg. 9399374)			
Functioning (single shot)	388 8-9 1/ 8/		4.5.6, 4.5.6.1, 4.5.6.2
1000 meter, min. Test	288	3.4	4.5.6, 4.5.6.1
Sensitivity Test	100 1-2 9/	3.4	4.5.6, 4.5.6.2
Rapid fire functioning test, 150 to 656.2 feet (200m)	100 3-4 1/	3.4	4.5.6, 4.5.6.3
Hot and cold temperature test	300 3/ 4/	3.4	4.5.6, 4.5.6.1, 4.5.6.2, 4.5.6.3
Petalling (missing petals)	200 5/ 1-2 6/	3.4	4.5.6, 4.5.6.2, 4.5.6.3
Petalling (blown base)	200 5/ 0-1	3.4	4.5.6, 4.5.6.2, 4.5.6.3

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:36Z
Check the source to verify that this is the current version before use.

TABLE II. First Article Inspection. - Continued

TITLE Projectile and Components (cont.)	SHEET 2 OF 2		DRAWING NUMBER See Below NEXT HIGHER ASSEMBLY
EXAMINATION OR TEST	N0. OF SAMPLE UNITS	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Safety	788 2/ 0-1	3.4.1	4.5.6, 4.5.6.1, 4.5.6.2, 4.5.6.3
Air Burst	788 2/ 1-2	3.4	4.5.6, 4.5.6.1, 4.5.6.2, 4.5.6.3
Jolt	12 0-1	3.6	4.5.13
Jumble	96 0-1	3.7	4.5.14
Transportation vibration	488	3.8	4.5.15
Non-functioning	100 3-4	3.3	4.5.7

Notes:
1/ Accept/reject numbers apply to cartridges that fail to provide all three of the impact signatures (flash, smoke and sound). For 1000 meter test, sound and smoke or sound and flash are also acceptable.
2/ Same projectiles that are subjected to 1000 meter min, Sensitivity Test and rapid fire tests.
3/ 100 rounds shall be subjected to each of the functioning tests listed above, see 4.5.6.
4/ Functioning test results except for air bursts and safety (3.4.1) are for informational purposes only.
5/ Projectiles fired in Sensitivity and rapid fire tests shall be recovered and inspected for missing petals and blown bases.
6/ One defect consists of one or more missing petals per projectile.
7/ If alternate knurling configuration is used, the projectile assemblies shall be subjected to temperature conditioned testing as specified by paragraph 4.5.10.
8/ This acceptance criteria applies for the total of the functioning failures (duds) from 1000 Meter Test and the Sensitivity Test.
9/ Accept/Reject numbers only apply to cartridges that fail to function on impact with the soft target, but function on impact behind the soft target.
10/ A sufficient number of assemblies shall be disassembled to obtain the sample specified in 4.5.8

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:36Z
Check the source to verify that this is the current version before use.

4.4 Conformance inspection.

4.4.1 Inspection lot formation. Lot formation shall be in accordance with MIL-STD-1916. For the components identified below, each inspection lot of projectiles shall be produced with components complying with the following constraints:

a. Projectile metal parts from one lot interfix number, from one manufacturer.
b. Escapments from no more than two consecutively produced serial lots from one manufacturer.
c. Flash charge composition from one lot number from one manufacturer.

4.4.2 Classification of characteristics.

a. Sampling requirements. Inspection sampling requirements for Critical, Major and Minor characteristics are defined in MIL-STD-1916. Unless specified otherwise, Inspection Level IV shall be used for all characteristics defined as Majors and Inspection Level II for all Minor characteristics; Critical characteristics shall be addressed in accordance with MIL-STD-1916.

b. Conformance inspection. Conformance inspection shall be performed in accordance with paragraph 4.4.2.1 through 4.4.2.19. For all conformance inspections the same sample specimen may be used for all non-destructive examinations or tests.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:36Z
Check the source to verify that this is the current version before use.

Conformance Inspection

Classification of Characteristics

PARAGRAPH	TITLE	SHEET 1 OF 1		DRAWING NUMBER 9399392
4.4.2.1	Pin, Firing			NEXT HIGHER ASSEMBLY 9399391
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
CRITICAL	None defined			
MAJOR				
101.	Profile of firing pin point	Level IV	3.2	Gage
102.	Length, firing pin point to backside of shoulder	Level IV	3.2	Gage
MINOR				
201.	Outside diameter at backside of shoulder	Level II	3.2	Gage
202.	Total length	Level II	3.2	Gage
203.	Evidence of poor workmanship	Level II	3.9	Visual/4.5.17
Notes:				

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:36Z
Check the source to verify that this is the current version before use.

Conformance Inspection

Classification of Characteristics

PARAGRAPH	TITLE	SHEET 1 OF 1		DRAWING NUMBER 9399393
4.4.2.2	Plate, Firing Plate			NEXT HIGHER ASSEMBLY 9399391
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
CRITICAL	None defined			
MAJOR				
101.	Diameter, Outside	Level IV	3.2	Gage
102.	True Position of center hole to outside diameter	Level IV	3.2	Gage
MINOR				
201.	Diameter, center hole	Level II	3.2	Gage
202.	Thickness	Level II	3.2	Gage
203.	Flatness	Level II	3.2	Gage
204.	Evidence of poor workmanship	Level II	3.9	Visual/4.5.17
Notes:				

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:36Z
Check the source to verify that this is the current version before use.

Conformance Inspection

Classification of Characteristics

PARAGRAPH	TITLE	SHEET 1 OF 1		DRAWING NUMBER 9399391
4.4.2.3	Firing Pin Assembly			NEXT HIGHER ASSEMBLY 9399390
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
CRITICAL	None defined			
MAJOR				
101.	Push test, firing pin assembly	Level III	3.2	4.5.1
MINOR				
201.	Evidence of poor workmanship	Level II	3.9	Visual/4.5.17
Notes:				

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:36Z
Check the source to verify that this is the current version before use.

Conformance Inspection

Classification of Characteristics

PARAGRAPH	TITLE	SHEET 1 OF 1		DRAWING NUMBER 9399394
4.4.2.4	Ogive, M918 TP			NEXT HIGHER ASSEMBLY 9399390
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
CRITICAL	None defined			
MAJOR				
101.	Major diameter, thread	Level IV	3.2	Gage
102.	Pitch diameter, thread	Level IV	3.2	Gage
103.	Largest outside diameter	Level IV	3.2	Gage
104.	Total length	Level IV	3.2	Gage
105.	Runout, largest outside diameter to thread pitch diameter	Level IV	3.2	Gage
106.	Runout, inside diameter to thread pitch diameter	Level IV	3.2	Gage
107.	Larger, inside diameter	Level IV	3.2	Gage
108.	Depth to step on inside diameter	Level IV	3.2	Gage
109.	Salt spray/protective finish	4 1/	3.2	4.5.2.1
110.	Missing or improperly placed groove	Level IV	3.2	Visual
MINOR				
201.	Length, full thread	Level II	3.2	Gage
202.	Length to shoulder at thread relief	Level II	3.2	Gage
203.	Diameter of thread relief	Level II	3.2	Gage
204.	Incorrect color	Level II	3.2	Visual
205.	Wall thickness	Level II	3.2	Gage
206.	Protective finish with bare spot exposing base metal except rack marks	Level II	3.8	Visual
207.	Evidence of poor workmanship	Level II	3.9	Visual/4.5.17

Notes:
1/ Rejection shall be in accordance with MIL-A-8625.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:36Z
Check the source to verify that this is the current version before use.

Conformance Inspection

Classification of Characteristics

PARAGRAPH	TITLE	SHEET 1 OF 1		DRAWING NUMBER 9399382
4.4.2.5	Insert, Body			NEXT HIGHER ASSEMBLY 9939374
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
CRITICAL	None defined			
MAJOR				
101.	Length, Body	Level IV	3.2	Gage
102.	Diameter outside	Level IV	3.2	Gage
103.	Parallelism of rear surface with front face	Level IV	3.2	Gage
104.	True position, setback spring cavity	Level IV	3.2	Gage
105.	True position, through hole to outside diameter	Level IV	3.2	Gage
106.	True position, powder cavity to outside diameter	Level IV	3.2	Gage
107.	Depth of escapement boss cavity	Level IV	3.2	Gage
MINOR				
201.	Diameter, powder cavity	Level II	3.2	Gage
202.	Depth, powder cavity	Level II	3.2	Gage
203.	Angle, outside diameter	Level II	3.2	Gage
204.	Location, angle	Level II	3.2	Gage
205.	Diameter, through hole	Level II	3.2	Gage
206.	Diameter, escapement boss cavity	Level II	3.2	Gage
207.	Evidence of poor workmanship	Level II	3.9	Visual/4.5.17

Notes:

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:36Z
Check the source to verify that this is the current version before use.

Conformance Inspection

Classification of Characteristics

PARAGRAPH	TITLE	SHEET 1 OF 1		DRAWING NUMBER 9399408 / NEXT HIGHER ASSEMBLY 9399390
4.4.2.6	Spring, Anti Creep			
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
CRITICAL	None defined			
MAJOR				
101.	Load test	Level III	3.2	4.5.3
102.	Length	Level IV	3.2	Gage
103.	Outer diameter	Level IV	3.2	Gage
MINOR				
201.	Evidence of poor workmanship	Level II	3.9	Visual/4.5.17

Notes:

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:36Z
Check the source to verify that this is the current version before use.

Conformance Inspection

Classification of Characteristics

PARAGRAPH	TITLE	SHEET 1 OF 1		DRAWING NUMBER 9399390
4.4.2.7	Ogive Assembly, 40mm, M918			NEXT HIGHER ASSEMBLY 9399374
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
CRITICAL				
1.	Anti creep spring missing 1/	100%	3.2	AAIE/4.5.11
MAJOR				
101.	Depth to firing pin point	Level IV	3.2	Gage
102.	Firing pin assembly improperly seated	Level IV	3.2	Visual
MINOR				
201.	Evidence of poor workmanship	Level II	3.9	Visual/4.5.17

Notes:
1/ Assembly shall be inspected in the inverted (open end of ogive facing down) position.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:36Z
Check the source to verify that this is the current version before use.

Conformance Inspection

Classification of Characteristics

PARAGRAPH	TITLE	SHEET 1 OF 1		DRAWING NUMBER 9399405
4.4.2.8	Cap, Inner			NEXT HIGHER ASSEMBLY 9399407
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
CRITICAL	None defined			
MAJOR	None defined			
MINOR				
201.	Length	Level II	3.2	Gage
202.	Outside diameter, max (closed end)	Level II	3.2	Gage
203.	Inside diameter, min	Level II	3.2	Gage
204.	Wall thick, min	Level II	3.2	Gage
205.	Evidence of poor workmanship	Level II	3.9	Visual/4.5.17

Notes:

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:36Z
Check the source to verify that this is the current version before use.

Conformance Inspection

Classification of Characteristics

PARAGRAPH	TITLE	SHEET 1 OF 1		DRAWING NUMBER 9399406
4.4.2.9	Cap, Outer			NEXT HIGHER ASSEMBLY 9399407
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
CRITICAL	None defined			
MAJOR	None defined			
MINOR				
201.	Length	Level II	3.2	Gage
202.	Outside diameter, max (closed end)	Level II	3.2	Gage
203.	Inside diameter, min	Level II	3.2	Gage
204.	Wall thick, min	Level II	3.2	Gage
205.	Evidence of poor workmanship	Level II	3.9	Visual/4.5.17

Notes:

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:36Z
Check the source to verify that this is the current version before use.

Conformance Inspection

Classification of Characteristics

PARAGRAPH	TITLE	SHEET 1 OF 1		DRAWING NUMBER 9399407
4.4.2.10	Capsule Assembly			NEXT HIGHER ASSEMBLY
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
CRITICAL	None defined			
MAJOR 101.	Weight of powder	100%	3.2	Gage
MINOR 201.	Evidence of poor workmanship	Level II	3.9	Visual/4.5.17
Notes:				

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:36Z
Check the source to verify that this is the current version before use.

Conformance Inspection

Classification of Characteristics

PARAGRAPH	TITLE	SHEET 1 OF 1		DRAWING NUMBER 9399396
4.4.2.11	Body			NEXT HIGHER ASSEMBLY 9399402
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
CRITICAL				
1.	Any perforation through the base	100%	3.2	AAIE/4.5.11
2.	Base thickness at center less than 0.009 inch	100%	3.2	AAIE/4.5.11
MAJOR				
101.	Large outside diameter	Level IV	3.2	Gage
102.	Runout at indicated lengths on body with pitch diameter (2)	Level IV	3.2	Gage
103.	Base thickness at center between 0.009 inch and 0.013 inch and above 0.020 inch	Level IV	3.2	Gage
104.	Thickness below groove (6) in base	Level IV	3.2	Gage
MINOR				
201.	Diameter of rotating band surface (after knurling)	Level II	3.2	Gage
202.	Width of rotating band surface	Level II	3.2	Gage
203.	Total length	Level II	3.2	Gage
204.	Length of effective thread, min	Level II	3.2	Gage
205.	Evidence of poor workmanship	Level II	3.9	Visual/4.5.17

Notes:

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:36Z
Check the source to verify that this is the current version before use.

Conformance Inspection

Classification of Characteristics

PARAGRAPH	TITLE	SHEET 1 OF 1		DRAWING NUMBER 9399402
4.4.2.12	Body assembly, prior to paint			NEXT HIGHER ASSEMBLY
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
CRITICAL	None defined			
MAJOR				
101.	Parallelism, back of rotating band with open end	Level IV	3.2	Gage
102.	Distance from open end to back of rotating band, max	Level IV	3.2	Gage
103.	Push test of rotating band	Level III	3.2	4.5.10
104.	Profile of rotating band	Level IV	3.2	Gage
105.	Runout of indicated point of taper with outside larger diameter	Level IV	3.2	Gage
106.	Location, large diameter on taper	Level IV	3.2	Gage
107.	Pitch diameter of thread	Level IV	3.2	Gage
108.	Minor diameter of thread	Level IV	3.2	Gage
MINOR				
201.	Evidence of poor workmanship	Level II	3.9	Visual/4.5.17

Notes:

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:36Z
Check the source to verify that this is the current version before use.

Conformance Inspection

Classification of Characteristics

PARAGRAPH	TITLE	SHEET 1 OF 1		DRAWING NUMBER 9399402
4.4.2.13	Body assembly, after painting			NEXT HIGHER ASSEMBLY 9399374
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
CRITICAL	None defined			
MAJOR				
101.	Body diameter max at boattail end	Level IV	3.2	Gage
102.	Diameter of rotating band	Level IV	3.2	Gage
103.	Salt spray/protective finish	1/	3.2	4.5.2.2
104.	Missing or improperly placed markings	Level IV	3.2	Visual
MINOR				
201.	Incorrect color of body	Level II	3.2	Visual
202.	Brown stripe missing or not 360 degrees	Level II	3.2	Visual
203.	Evidence of poor workmanship	Level II	3.9	Visual/4.5.17

Notes:
1/ Sampling and rejection shall be in accordance with TT-C-490.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:36Z
Check the source to verify that this is the current version before use.

Conformance Inspection

Classification of Characteristics

PARAGRAPH	TITLE	SHEET 1 OF 2		DRAWING NUMBER 9399374
4.4.2.14	Projectile Assembly			NEXT HIGHER ASSEMBLY 9399372
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
CRITICAL				
1.	Rotor in armed or partially armed position 1/ 6/	100%	3.2	AAIE/4.5.11
2.	Detent Spring Missing 1/	100%	3.2	AAIE/4.5.11
3.	Detent Missing 1/	100%	3.2	AAIE/4.5.11
4.	Detent not engaged in teeth of sector gear 1/ 7/	100%	3.2	AAIE/4.5.11
5.	Setback spring missing 1/	100%	3.2	AAIE/4.5.11
6.	Setback pin missing or retracted 1/ 8/	100%	3.2	AAIE/4.5.11
7.	Safety	244 2/ 0-1	3.4.1	4.5.6, 4.5.6.1, 4.5.6.2
8.	Jolt 11/	12 0-1	3.6	4.5.13
9.	Jumble 11/	96 0-1	3.7	4.5.14
10.	Non-Functioning	10 0-1	3.3	4.5.7
11.	Transportation vibration	204	3.8	4.5.15
MAJOR				
101.	Total Length, max	Level IV	3.2	Gage
102.	Runout, rotating band to ogive diameter	Level IV	3.2	Gage
103.	Assembly torque, prior to sealant cure	3/	3.2	4.5.4
104.	Pressure test	125 3-4	3.2	4.5.5
105.	Functioning	194 9/ 5-6	3.4	4.5.6, 4.5.6.1, 4.5.6.2
	1000 meter, min, Impact test	144	3.4	4.5.6, 4.5.6.1
	Sensitivity Test	50 1-2 10/	3.4	4.5.6, 4.5.6.2
106.	Early burst	244 2/ 0-1	3.4	4.5.6, 4.5.6.1, 4.5.6.2
107.	Petalling (Missing Petals)	50 4/ 1-2 5/	3.4	4.5.6, 4.5.6.2
108.	Petalling (Blown Base)	50 4/ 0-1	3.4	4.5.6, 4.5.6.2

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:36Z
Check the source to verify that this is the current version before use.

Conformance Inspection

Classification of Characteristics

PARAGRAPH	TITLE	SHEET 2 OF 2		DRAWING NUMBER 9399374
4.4.2.14	Projectile Assembly (cont.)			NEXT HIGHER ASSEMBLY 9399372
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
MINOR				
201.	Markings missing, improper or illegible	Level II	3.2	Visual
202.	Evidence of poor workmanship	Level II	3.9	Visual/4.5.17

Notes:
1/ Performed on escapement assembly prior to assembly to projectile.
2/ Same projectiles that are subjected to non-functioning, 1000 meter Impact test and Sensitivity Test.
3/ Table I (with verification level (VL) II and Table IV (Continuous Sampling Plan) of MIL-STD-1916 are applicable.
4/ Projectile fired in 135 to 200 feet shall be recovered and inspected for missing petals and blown bases.
5/ One defect consists of one or more missing petals per projectile.
6/ A partially armed position is defined as a rotor assembly with two (2) or more teeth in the armed position. The full armed position is when the detonator is centered in the spitback hole.
7/ Either the protrusion of the detent into the sector gear teeth or the seating of the detent in its forward position shall be inspected.
8/ A retracted setback pin is a setback pin in an escapement assembly with the dimension (excluding cutoff nib) from the top of it to the bottom of the escapement housing. Dwg. 8886358, being 0.040 or greater.
9/ Sample size reflects the rounds for the Sensitivity Test and for the 1000 meter impact test. This acceptance criteria applies for the total of the functioning failures (duds) and cartridges that do not provide flash, smoke or sound, from 1000M Test and the Sensitivity Test. For 1,000 meter test, sound and smoke or sound and flash are also acceptable.
10/ Accept/Reject numbers apply only to cartridges that fail to function on impact with the soft target, but function on impact behind the soft target.
11/ If a detonator initiates during this test, the failure shall be considered a “Major” defect if the fuze is in the unarmed position.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:36Z
Check the source to verify that this is the current version before use.

Conformance Inspection

Classification of Characteristics

PARAGRAPH	TITLE	SHEET 1 OF 1		DRAWING NUMBER 9399389
4.4.2.15	Flash Charge Composition, bulk, prior to loading into capsule			NEXT HIGHER ASSEMBLY 9399407
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
CRITICAL	None defined			
MAJOR 101.	Moisture Content	Per batch 1/	3.2	4.5.8
MINOR 201.	Evidence of poor workmanship	Level II	3.9	Visual/4.5.17

Notes:
1/ If the batch of flash charge composition fails the requirement of the drawing, it shall be classed defective and not used in production.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:36Z
Check the source to verify that this is the current version before use.

Conformance Inspection

Classification of Characteristics

PARAGRAPH	TITLE	SHEET 1 OF 1		DRAWING NUMBER 12624588
4.4.2.16	Unsealed inner fiberboard box			NEXT HIGHER ASSEMBLY
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
CRITICAL	None defined			
MAJOR				
101.	Improper number of projectiles	Level IV	3.2	Visual
102.	Projectile Inverted	Level IV	3.2	Visual
103.	Filler missing	Level IV	3.2	Visual
104.	Supports missing	Level IV	3.2	Visual
MINOR				
201.	Evidence of poor workmanship	Level II	3.9	Visual/4.5.17

Notes:

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:36Z
Check the source to verify that this is the current version before use.

Conformance Inspection

Classification of Characteristics

PARAGRAPH	TITLE	SHEET 1 OF 1		DRAWING NUMBER 12624588
4.4.2.17	Sealed Inner Fiberboard Box, Prior to Sealing Bag			NEXT HIGHER ASSEMBLY
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
CRITICAL	None defined			
MAJOR				
101.	Container damaged	Level IV	3.2	Visual
102.	Corners not blunted	Level IV	3.2	Visual
MINOR				
201.	Evidence of poor workmanship	Level II	3.9	Visual/4.5.17

Notes:

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:36Z
Check the source to verify that this is the current version before use.

Conformance Inspection

Classification of Characteristics

PARAGRAPH	TITLE	SHEET 1 OF 1		DRAWING NUMBER 12624588
4.4.2.18	Unsealed Outer Box			NEXT HIGHER ASSEMBLY
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
CRITICAL	None defined			
MAJOR				
101.	Heat seal test of packing seals (after sealing bag)	Level IV	3.2	4.5.9
102.	Bag torn or perforated	Level IV	3.2	Visual
103.	Bag improperly sealed	Level IV	3.2	Visual
104.	Fillers missing	Level IV	3.2	Visual
MINOR				
201.	Evidence of poor workmanship	Level II	3.9	Visual/4.5.17

Notes:

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:36Z
Check the source to verify that this is the current version before use.

Conformance Inspection

Classification of Characteristics

PARAGRAPH	TITLE	SHEET 1 OF 1		DRAWING NUMBER 12624588
4.4.2.19	Sealed Outer Box			NEXT HIGHER ASSEMBLY
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
CRITICAL	None defined			
MAJOR				
101.	Boxed damaged	Level IV	3.2	Visual
MINOR				
201.	Marking missing or unidentifiable	Level II	3.2	Visual
202.	DOT marking missing	Level II	3.2	Visual
203.	Evidence of poor workmanship	Level II	3.9	Visual/4.5.17

Notes:

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:36Z
Check the source to verify that this is the current version before use.

4.5 Methods of inspection.

4.5.1 Push test of firing pin assembly. The firing pin assembly shall be placed in an approved fixture and the push test specified on Dwg. 9399391 shall be applied. This test is a non-destructive test. Parts successfully tested may be returned to the lot. If the part fails to meet the requirement the lot shall be rejected and set aside for disposition in accordance with MIL-STD-1916

4.5.2 Salt spray.

4.5.2.1 Ogive, M918 TP. This test shall be conducted in accordance with ASTM B117 for 48 hours. Salt solution shall be 5 percent. This test is a destructive test. Parts so tested shall not be returned to the lot.

4.5.2.2 Body Assembly. This test shall be conducted in accordance with ASTM B117. This test shall be conducted for 48 hours along 1.25 inches of the outside length measured from the open end. The surface area rear of this point may be tested for 24 hours minimum. Salt spray test not required for inside surface of body assembly. The rotating band may be masked off for this test. Salt solution shall be 5 percent. This test is a destructive test. Parts so tested may not be returned to the lot.

4.5.3 Load test of anti-creep spring. The anti-creep spring shall be placed in a fixture and the load specified on Dwg. 9399408 shall be applied. This test is a non-destructive-test. Parts successfully tested may be returned to the lot.

4.5.4 Assembly torque of ogive assembly prior to sealant cure. The projectile assembly shall be placed in an approved fixture and the torque specified on the applicable drawing shall be applied. Observation shall be made for the applicable requirement. This test is a non-destructive test. Parts successfully tested may be returned to the lot.

4.5.5 Air Pressure test. The projectile shall be placed in an approved fixture and air applied to produce the pressure specified on the applicable drawing. Observations shall be made for a drop in pressure indicating a leak. This is a non-destructive test. Parts successfully tested may be returned to the lot.

4.5.6 Functioning. This test shall be conducted as follows:

After completion of the transportation-vibration test, the projectile assemblies, being tested shall be assembled as complete cartridges, (Dwg. 9399372) temperature conditioned at 70° ± 5°F for at least eight (8) hours, and then fired within fifteen (15) minutes after being removed from temperature soak. During first article inspection only, hot and cold temperature testing shall be performed for the tests specified in 4.5.6.1, 4.5.6.2 and 4.5.6.3. For the high temperature test, the cartridges shall be conditioned at 125°F ± 5°F for at least four (4) hours but no more than twelve (12) hours. For the low temperature test, the cartridges shall be conditioned at -25°F ± 5°F for at least four (4) hours. Fifty (50) cartridges shall be fired for each hot and each cold portion of each of the three (3) tests. During hot and cold temperature testing only failure to comply with the safety (3.4.1), premature, and early burst requirements shall be cause for rejection of the first article quantity. Functioning during hot and cold temperature

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:36Z
Check the source to verify that this is the current version before use.

testing shall be measured for informational purposes only. Test cartridges shall be fired from an approved weapon. All tests performed shall be observed from the gun position.

4.5.6.1 1,000 meters, minimum Impact Test. After completion of the transportation-vibration test, the cartridges shall be fired and impact at 1000 meter minimum from the weapon on level terrain. Observation shall be made for gun stoppages (information only), projectile stuck in bore, premature bursts, early bursts, and functioning on the impact medium.

4.5.6.2 Sensitivity Test. This test shall be performed to verify the sensitivity of the projectile assembly when striking a soft target.

For this test, a sensitivity failure is defined as a round that fails to function on impact with the soft target, but functions on impact behind the soft target.

If the projectile assembly fails to function on both the soft target and the secondary impact medium behind the soft target it shall be counted as a functioning failure, but not as a sensitivity failure.

A functioning failure from this test shall be added to the functioning failures, if any, from the 1,000 meters impact test. The acceptance criteria for functioning shall be applied to the combined sample.

The test shall be performed as follows:

After completion of the transportation-vibration test, the cartridges shall be fired and impact at 135 to 200 ft. from the machine gun on a.5 inch cellulosic fiber insulating board per ASTM C208 or approved equal target material. A secondary impact medium to allow recovery shall be a minimum of 20 ft. behind the fiberboard target. Recovered projectiles shall be inspected for missing petals or blown base. Functioning must be taped with a video camera with audio capability located along the side of the soft target. The camera does not need to be a high speed camera. The camera must be facing the impact area in such a way that allows to easily determining whether functioning occurred on the soft target or on the secondary impact medium. Observation shall be made for gun stoppages (information only), projectile stuck in bore, premature bursts, early bursts and functioning on impact.

4.5.6.3 Rapid fire functioning. The linked cartridges shall be fired in bursts of five (5), and impact at 150 to 656.2 feet (200M) from the MK 19, MOD III, Grenade Machinegun into 10 inches minimum of coarse sand graded at approximately 25° with the horizontal. The sand shall be raked so that individual craters do not overlap. In addition, the sand shall be raked after each function failure. If a projectile assembly falls outside the impact medium, it shall be declared a "no-test". When reporting ballistic test results, the number of "no-test" cartridges shall also be reported. Observation shall be made for gun stoppages (information only), projectile stuck in bore, premature bursts, early bursts, and functioning on the impact medium. This test shall be preformed at an installation approved by the government.

4.5.7 Non-functioning. After completion of the transportation-vibration test, the projectile assemblies shall be removed from their regular shipping containers and subjected to this test. The Projectile Assembly shall be assembled to a M169, 40MM Cartridge Case Assembly (Dwg. 8886327) resulting in a M918 TP Cartridge (Dwg. 9399372). The cartridges shall be fired from the launcher against a vertical soft target, approved by the Contracting Officer. The target shall be placed 20 feet plus or minus

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:36Z
Check the source to verify that this is the current version before use.

two (2) foot from point of fire. The projectile velocity shall be measured and shall comply with the requirements of 790 ± 10 feet per second (fps) and a standard deviation not exceeding 12.0 fps. Prior to FAT and each LAT, one test shot shall be fired at the proposed target to check for detonation at a range of 100 meters to proof test target. If one or more projectiles function on the target or a premature burst occurs the lot shall be rejected and set aside for disposition in accordance with MIL-STD-1916. The subsequent function or non-function of the fuze down range of the target shall be measured.

4.5.8 Moisture content of flash charge. An accurately weighed position of three (3) grams (g.) of the charge composition shall be transferred to a tared dish. The moisture dish and contents shall be placed in a drying oven at 70 ± 2°C for 4 hours, cooled in a desiccator and weighed. The percentage of moisture in the sample shall be calculated as follows:

$$\text{Percentage Moisture} = \frac{100A}{W}$$

Where:

A= Loss in weight of moisture dish and contents.

W = Weight of sample, g.

4.5.9 Heat seal test of packing seals. This test shall be performed in accordance with specification MIL-STD-2073-1.

4.5.10 Push test of rotating band. The body assembly shall be placed in an approved fixture and the force specified on the applicable drawing shall be applied. Observation shall be made for compliance with the applicable requirement. This is a non-destructive test. Parts so tested may be returned to the lot.

Every time there is a change in the knurling configuration, a First Article Test shall be performed. The body assemblies shall be subjected to temperature conditioning testing. For the ambient portion of the test the bodies shall be conditioned at 70° ± 5°F for at least eight (8) hours, and then tested within fifteen (15) minutes after being removed from the temperature soak. For the high temperature test, the body assemblies shall be conditioned at 125° ± 5°F for at least four (4) hours but no more than twelve (12) hours, and then tested within fifteen (15) minutes after being removed from the temperature soak. For the low temperature test, the body assemblies shall be conditioned at -25° ± 5°F for at least four (4) hours, and then tested within fifteen (15) minutes after being removed from the temperature soak. Fifty (50) body assemblies shall be tested for each the hot, cold and ambient portions of this test. Failure to meet the requirement on the specified drawing shall be reason for rejection of the First Article.

4.5.11 Inspection Equipment. The inspection equipment required to perform the examination and tests prescribed herein is described in the Inspection Method Column in the tables starting with paragraph 4.4.2.1. Inspection equipment used for acceptance of product shall be capable of repeatable measurements, by various experienced inspection/test personnel, to an accuracy of 10% or better of the total tolerance of the characteristic being inspected. Any automated acceptance inspection equipment (AAIE)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:36Z
Check the source to verify that this is the current version before use.

prove-out shall be performed in accordance with MIL-A-70625 (see 6.9).

4.5.12 Lot identification. Visually verify that each lot of projectiles is identified as the type, caliber, model and lot number in accordance with MIL-STD-1168.

4.5.13 Jolt. Prior to the jolt test, the escapement assemblies, assembled in accordance with projectile assembly Dwg. 9399374 (except without flash charge composition), shall be x-rayed in accordance with 4.5.16 to determine compliance with 3.6. They shall then be tested in accordance with Test A1-1 of MIL-STD-331B. After completion of the test, the projectile assemblies shall be x-rayed again in accordance with 4.5.16 to determine compliance with 3.6. This is a destructive test. Assemblies so tested shall not be returned to the lot.

4.5.14 Jumble. Prior to jumble test, the escapement assemblies assembled in accordance with projectile assembly Dwg. 9399374, (except without flash charge composition), shall be x-rayed in accordance with 4.5.16 to determine compliance with 3.7. They shall then be tested in accordance with Test A2-1.of MIL-STD-331B.. After completion of the test, the projectile assemblies shall be x-rayed again in accordance with 4.5.16 to determine compliance with 3.7. This is a destructive test. Assemblies so tested shall not be returned to the lot.

4.5.15 Transportation vibration. Prior to transportation vibration test the projectile assemblies shall be x-rayed in accordance with 4.5.16. The projectiles shall then be packed in their regular shipping containers and be tested in accordance with MIL-STD-331 (Test No.B2, with vibration levels conforming to 6.1.2.1 of Test B1). The test shall be performed at ambient temperature only. Simulated projectiles may be used to insure a full box. After completion of the transportation vibration test the projectiles shall be x-rayed again in accordance with 4.5.16 to determine compliance with 3.3. If one or more fuzes fail to meet the requirements of 3.3 or has experienced damage that compromises any safety feature of the projectile the lot shall be rejected and set aside for disposition in accordance with MIL-STD-1916. Parts subjected to subsequent tests shall not be returned to the lot.

4.5.16 X-ray of fuze assembly. Examination shall be performed with X-ray equipment or any other method acceptable to the contracting officer. Radioscopic examination equipment, procedures, and personnel shall be in accordance with the requirements of ASTM E1255. The fuze assembly shall be examined for compliance with 3.2, 3.3, 3.4, or 3.5 as applicable. The test shall be conducted in two (2) planes:

a. From the top of the fuze assembly down.
b. From the side

4.5.17 Workmanship. Visually verify that all parts and assemblies meet requirements of paragraph 3.6.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:36Z
Check the source to verify that this is the current version before use.

5. PACKAGING

For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This specification covers the metal parts and loading, assembling and packaging for Projectile assembly for Cartridge, 40MM: M918 TP.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. Title, number (w/revision level) and date of all reference documents.
c. Quantity required and delivery schedule.
d. Lists of drawings, publications and specifications, showing applicable revision dates.
e. Responsibility for furnishing test machine gun, cartridge case assemblies, escapement assemblies and links.
f. Criteria for inspection of parts and assemblies.
g. Packaging instructions, if different.
h. Quality Conformance Inspection, if other than specified in Section 4 of this specification.
i. First Article Sample requirements, other than specified in Section 4 of this specification.
j. Critical Inspection Equipment (see 6.9)

6.3 Submission of results of contractor conducted examinations and tests. Unless otherwise specified by the Contracting Officer, the contractor should forward requested records of examination or tests to the Contracting officer **.**

6.4 Submission of test data. In addition to the normal distribution of records, when the cartridges are procured by JMC, one (1) copy of all ballistic data and ammunition data cards should be forwarded to the Contracting officer .

6.5 First article. When a first article inspection is required, the contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, test and approval of the documents for first article.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:36Z
Check the source to verify that this is the current version before use.

6.6 Definitions.

6.6.1 Early burst. A projectile that functions in flight further than the arming distance from the weapon muzzle.

6.6.2 Premature burst. A premature burst is considered to have occurred if the projectile functions in the weapon bore or in flight up to the arming distance from the weapon muzzle.

6.6.3 Sensitivity Failure. A sensitivity failure is defined as a round that, during the Sensitivity Test fails to function on impact with the soft target but functions on impact behind the soft target.

6.6.4 Function failure. A function failure is defined as round that fails to function after being fired. During sensitivity test, a function failure is defined as a round that fails to function on impact with the soft target and does not function upon impact behind the soft target.

6.7 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research, Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, Picatinny Arsenal, US Army Armament Research and Development Command (ARRADCOM) or US Army Armament Research and Development Center (ARDC) drawings. Technical data originally prepared by these activities are now under the cognizance of ARDEC.

6.8 Subject term (key word) listing.

M918 TP Cartridge
MK19, MOD III, Grenade Launcher
M550 Escapement Assembly

6.9 Critical Inspection Equipment.

6.9.1 Submission of inspection equipment designs for approval. See MIL-A-70625. Submit equipment designs as required to Commander, US Army ARDEC, AMSRD-AAR-QEM-D, Picatinny, NJ 07806-5000. Request letter of submittal state contractor, contract number, specification number, item nomenclature and classification of defects or test paragraph.

6.9.2 System Reliability. The contractor should provide for each defect identified as Critical, a manufacturing and inspection system that assured no more than one in a million contain the defect. This should be assured by controlling the maximum defect rate produced and the error rate of the inspection equipment such that the product of the two terms is less than one in a million.

6.9.3 Maximum defect rate produced. The maximum defect rate produced should be defined as the largest defect rate expected for the characteristics of concern on a

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:36Z
Check the source to verify that this is the current version before use.

monthly or lot basis. The contractor should establish the maximum defect rate, however, it may not exceed one percent without approval of the procuring contracting officer. Once established, the contractor should monitor the defect rate to assure it does not exceed the maximum rate allotted. If the established maximum defect rate is exceeded, the production of that feature should be terminated until the cause is determined and corrected. Additionally, all products for that period of time should be rejected and re-inspected.

6.9.4 Maximum error rate of the inspection system. The error rate of the inspection equipment should be defined as the expected ratio of the number of defective parts accepted to the number of defective parts inspected by the equipment. The contractor, to meet the system reliability as defined in contract, defines the maximum error rate allowed. However, it may not exceed 1/500 without approval of the procuring contracting officer. Based on the maximum error rate defined for the equipment, the contractor should develop demonstration test procedure to demonstrate the error rate of the equipment. The test should be performed using defective parts or reject standards. No part or standard should be accepted during the test. If a part or standard is accepted, the cause for failure should be isolated and the corrected and the test rerun. Unless otherwise specified in the detailed item specification, the minimum number of test samples to be run should be equal to 0.7 divided by the error rate (e.g. If the required error rate is 1/1000, the sample size would be 0.7 divided by (1/1000) = 700.)

6.9.5 Periodic verification. Periodic verification of the system error rate should be performed.

6.9.6 Redundant inspection equipment. The contractor may elect to use redundant inspection equipment in lieu of using a single piece of equipment with a very low error rate. Where redundant equipment is used, the inspection system error rate is defined as the error rates of the multiple pieces of equipment multiplied together.

6.10 Standard deviation should be calculated from the following formula:

$$S = \sqrt{\frac{\sum (x - \bar{x})^2}{n - 1}}$$

Where: x = each individual value

$\bar{x}$ = sample arithmetic mean $\quad \frac{\sum x}{n}$

n = sample size

$\sum (x - \bar{x})^2$ = The sum of the squares of the differences between the sample mean and each individual value.

6.10.1 Alternate method. For a faster and easier method of arriving at the standard deviation formula, the following may be used:

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:36Z
Check the source to verify that this is the current version before use.

$$S = \sqrt{\frac{n\sum x^2 - (\sum x)^2}{n(n-1)}}$$

6.11 Revisions/amendments history. The following is a record of changes incorporated into this document

ECP	Description	Date Approved
R04Q2019	Initial Release of M918 Projectile Assembly MIL-DTL specification	Aug 2004

6.12 Interim Revision This interim revision has been prepared at Program Manager request to document in ASSIST Online the version of a draft specification that was placed on contract in 2004. This interim revision has been dated to match the date of the draft specification that was placed on contract. It has been modified slightly to comply with MIL-STD-961 requirements for format, but not content. A fully coordinated revision, compliant with both MIL-STD-961 format and content requirements, will be approved for publication.

6.13 Changes from previous issues. Marginal notations are not used in this revision to identify changes with respect to the previous issues due to the extent of the changes.

Preparing activity:
Army-AR
(Project No. 1310-2006-009)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:36Z
Check the source to verify that this is the current version before use.